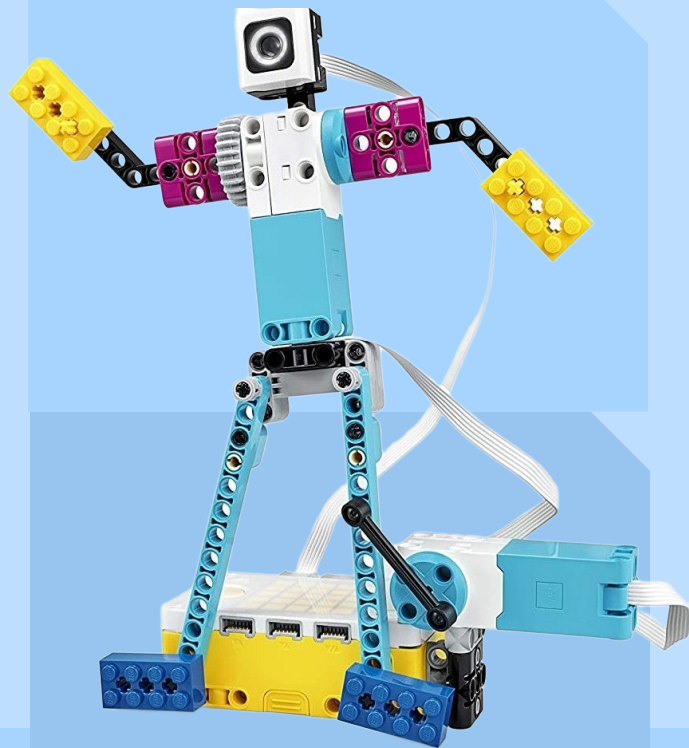


LESSON 2

Intro to Spike Prime



Spike Prime Materials

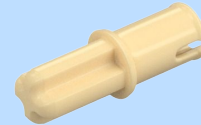
1 Pin
2 Types
Friction/Frictionless(Grey)



2 Pin
2 Types
Friction/Frictionless(Beige)

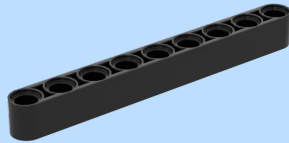


Pin/Axle Connector
2 Types
Friction/Frictionless(Beige)

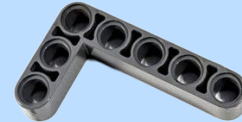


Spike Prime Materials

Beam
1 X (number of holes)
Only odd number of holes

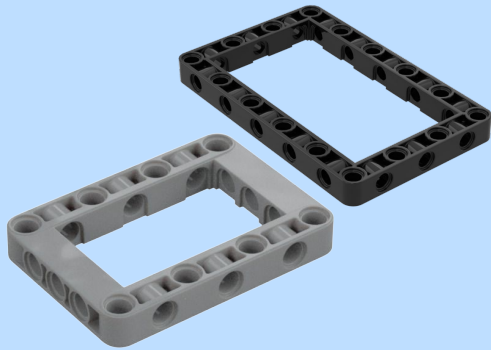


Angled Beam
90,45,30

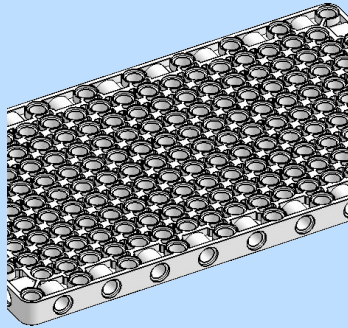


Spike Prime Materials

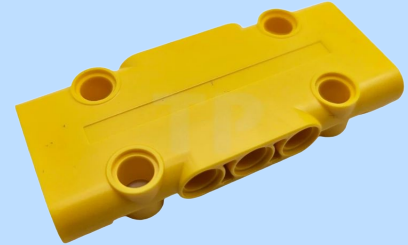
Picture Frame
Small, Med, Large



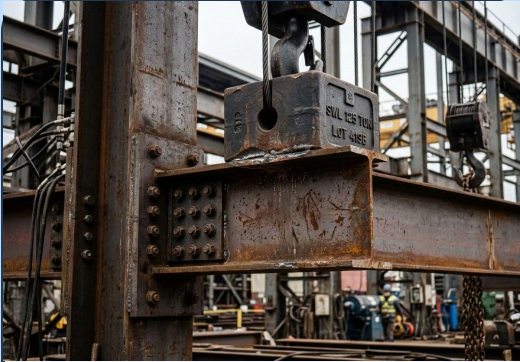
Panel
11x19



Plates
Small, Med, Large



Engineering Words for Today



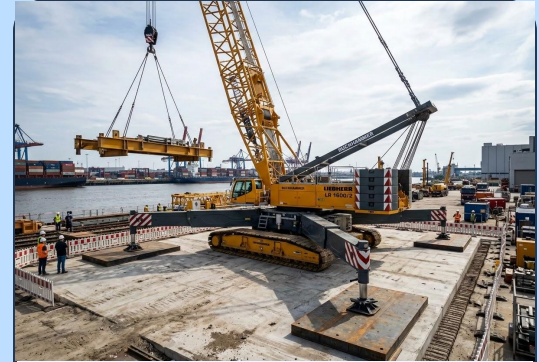
Strong

Can hold weight or force without breaking.



Rigid

Does not bend, wiggle, or twist easily.



Stable

Does not fall over easily.

Structural Forces: Tension & Compression

The forces that act on members are called tensile forces or compression forces.

Tensile Forces

These forces will **stretch** the structure.



Compression Forces

These forces will **squeeze** the structure.

Defining Structural Members



Ties

Members that are in **tension**.

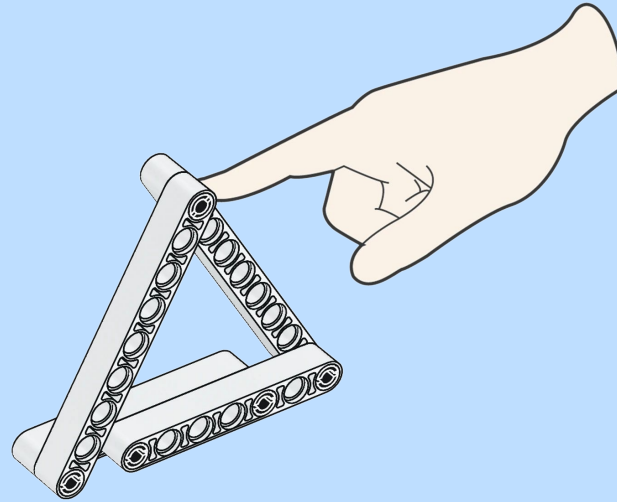
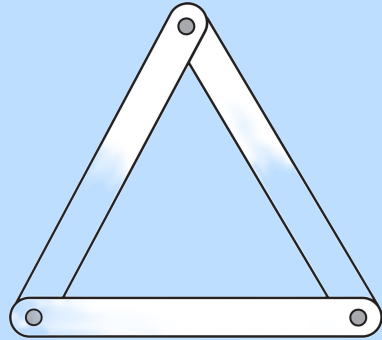


Struts

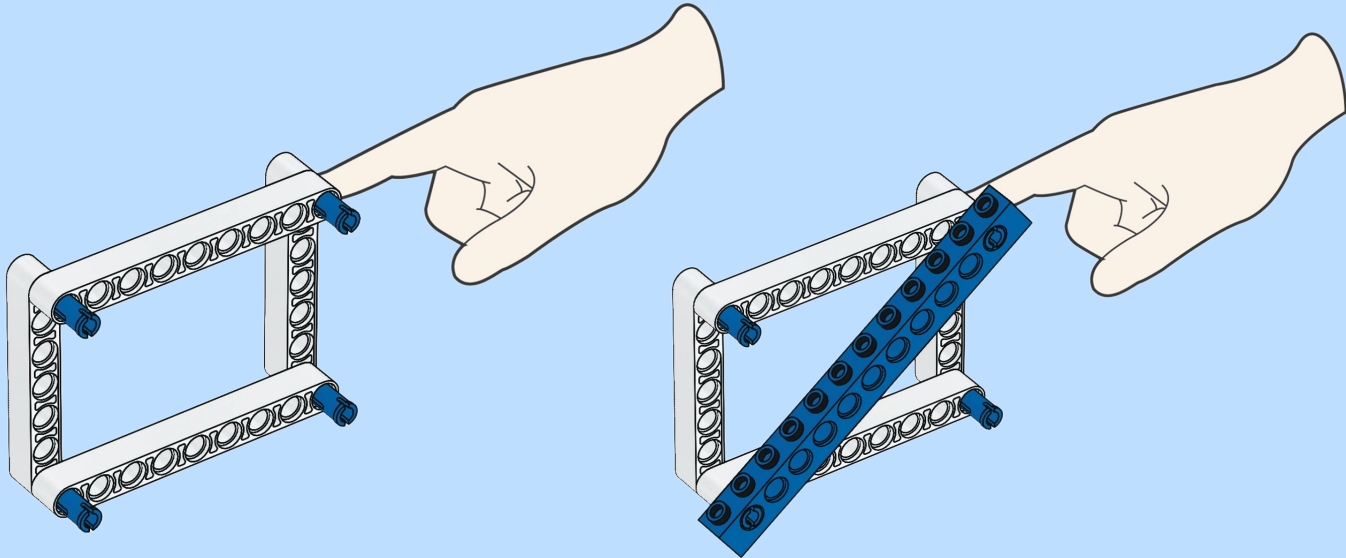
Members that are under **compression**.

Building Strong Structures Tip 2

In bridges, cranes, towers and even space stations, triangulation is often used to make structures rigid.



A frame structure is made from pieces called members. This frame is rigid because it is triangulated.



Today's Challenge: Building a Bridge

- Create a bridge that is 14 inches apart
- Bridges will be tested as such:
 - Cup will be placed
 - Marbles will be placed inside the cups
- When the timer goes off please raise your hand